Detection of Multi-Modal Deepfakes

A Systematic Comparison of Architectures for Audio-Visual Video-Level Classification
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Abstract

This thesis addresses the detection of multimodal deepfakes at the video level. It investigates how different architectures for combining visual and acoustic information behave in terms of detection performance and generalization across datasets. Three phases are developed and compared under identical conditions. Phase 1 combines a ResNet-50 with a BiLSTM, Phase 2 uses X3D-M, and Phase 3 employs TimeSformer with a cross-attention fusion. The audio branch, based on HuBERT, remains constant across all phases. The models are trained in a binary and, where the datasets allow, in a four-class formulation on AV-Deepfake1M++, FakeAVCeleb and DeepSpeak v2.0, and are evaluated across all dataset combinations.

The results show that the pipeline works reliably in-domain, while generalization across datasets is limited. The audio branch generalizes better than the video branch, whose cross-dataset performance drops to chance level under uniform frame sampling. This effect is shown to depend on the sampling strategy and can be reduced through segment-based training. The increasing architectural complexity does not lead to better performance, as Phase 3 falls behind the simpler phases both in-domain and cross-dataset.

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1. Introduction

In recent years, the proliferation of manipulated video content has grown significantly. Deepfakes refer to artificial intelligence-driven manipulations in which existing recordings are deliberately altered. A particular form is so-called face-hopping or face-swapping, in which the face of one person in an existing video or image is replaced with the face of another person. Unlike the complete synthesis of new video content, authentic material is modified in this case, whereby the forgery assumes the context and credibility of the original recording. Such manipulations of existing videos, which can affect both the image and audio tracks, form the research subject of the present thesis. Through open-source implementations and freely available tools, the technical barriers to creating such forgeries have been lowered. Whereas specialized knowledge and expensive hardware were previously required, nowadays individuals with less technical expertise can create convincing forgeries.

Particularly in the context of misinformation and fake news, such forgeries present a challenge for digital forensics. If a person's face is inserted into a foreign video or their voice is replicated, it is possible to attribute to them statements and actions that never took place. Deepfakes have already been misused in various contexts, from political misinformation to fabricated evidence to identity theft. Since viewers can no longer easily determine whether a recording is authentic, such forgeries can undermine public trust in video material as a whole. Therefore, it is necessary to develop effective detection methods.

1.1. Problem Statement

A large portion of existing detection procedures considers only the visual modality and makes a binary decision about whether a video is authentic or manipulated. However, modern manipulations often affect only one of the two modalities. For example, the audio track of a video can be altered through synthetic speech synthesis while the image remains unchanged, or vice versa. A purely visual procedure cannot detect a pure audio manipulation, since the image is authentic. Such cases can only be detected if both modalities are considered together.

Binary classification also provides no information about which modality is affected. For forensic assessment of a video, this distinction is relevant, as audio and video manipulations differ in their consequences and further analysis. A classification that distinguishes between authentic material, manipulated audio track, manipulated image track, and doubly manipulated material provides a more accurate picture than a simple yes-no decision.

Another problem concerns generalization. Detection models are usually trained and evaluated on a single dataset. In this constellation, they achieve high detection rates. However, when the same models are tested on a different dataset with different manipulation techniques, performance drops noticeably. This cross-dataset generalization is crucial for practical deployment, since real manipulations rarely match the training data. However, it is less frequently investigated in the literature than pure in-domain performance.

Finally, there is a lack of systematic comparison of different architectural approaches under identical conditions. Numerous works present individual systems, but a direct comparison of different video encoders and fusion strategies on the same data and with the same training procedure is lacking. Without such a comparison, it is difficult to assess which components actually contribute to detection performance.

1.2. Objectives and Approach

The goal of this thesis is to develop a multimodal system for video-level detection of deepfakes and to systematically compare different architectural variants. Both binary classification (Real vs. Fake) and four-class classification are considered. The four classes differ in the affected modality: Real (R), Fake Audio / Real Visual (FARV), Real Audio / Fake Visual (RAFV), and Fake Audio / Fake Visual (FAFV).

The approach is based on three progressive architectures, referred to hereafter as phases. All three phases use HuBERT [12] as an audio encoder, but differ in the video encoder and fusion strategy. Phase 1 combines ResNet-50 with a bidirectional LSTM for temporal modeling and fuses the modalities through concatenation. Phase 2 replaces the video branch with the 3D-CNN X3D-M, which processes spatial and temporal information together. Phase 3 uses TimeSformer as a video encoder and replaces concatenation with cross-attention fusion. This progression allows investigation of the influence of increasingly complex video encoders and fusion strategies on detection performance.

Evaluation is conducted on three datasets: AV-Deepfake1M++ [6], FakeAVCeleb, and DeepSpeak v2.0 [4]. All trained models are evaluated not only in-domain but also cross-dataset by testing each model on the other datasets. Additionally, performance is analyzed separately by modality to assess the contribution of each modality individually.

1.3. Contributions

This thesis makes the following contributions to multimodal deepfake detection:

1. **Systematic Architectural Comparison:** Three architectures with increasingly complex video encoders and fusion strategies are compared under identical training and evaluation conditions. This isolates the influence of individual components on detection performance.
2. **Modality-Aware Four-Class Classification:** In addition to binary classification, a four-class formulation is investigated that indicates which modality was manipulated. This distinction is practically relevant for deepfake forensics.
3. **Cross-Dataset Generalization:** All models are evaluated across all dataset combinations. The analysis reveals a pronounced asymmetry between modalities that is relevant for understanding the generalization capabilities of multimodal systems.
4. **Modality-Separated Evaluation:** By separately evaluating the audio, video, and fusion branches, the contribution of each modality is individually quantified. It shows that fusion can be impaired by a poorly generalizing video branch.

1.4. Structure of the Thesis

The thesis is structured as follows. Chapter 2 explains the fundamental concepts necessary for understanding the thesis. This includes descriptions of different types of deepfakes as well as the technical foundations of the architectures used, in particular self-supervised learning models for audio, convolutional networks, transformers, and fusion strategies.

Chapter 3 analyzes existing approaches to deepfake detection. The focus is on multimodal methods for video-level classification. Approaches for temporal localization are presented as a contrasting research direction. At the end, the research gap addressed by this thesis is identified.

Chapter 4 describes the developed system in detail. The architectures of the three phases are explained and the associated design decisions are justified. The description includes the preprocessing pipeline, feature extraction through video and audio encoders, and fusion strategies.

Chapter 5 presents the experimental results. The evaluation includes in-domain performance per phase, cross-dataset generalization across all dataset combinations, and a modality-separated analysis whose central findings are contextualized in Section 5.7.

Chapter 6 summarizes the key findings and discusses future research directions.

2. Background

This chapter explains the fundamental concepts necessary for understanding the present thesis. First, the different types of deepfake manipulations are described, as these form the basis of the detection task. Subsequently, the architectures used for video and audio analysis, the fusion strategies employed, and the evaluation metrics used are explained.

2.1. Deepfake Manipulation Types

The term *deepfake* refers to AI-generated or AI-manipulated media content that imitates or falsifies authentic content. In the video domain, various types of manipulation can be distinguished that leave different artifacts and require different detection strategies.

2.1.1. Face Swapping

Face swapping refers to the exchange of a person's face in a video with the face of another person. Here, facial expressions, head movements, and the context of the target video are preserved, while the identity is exchanged. Modern methods such as SimSwap or FaceShifter use GAN-based architectures (Generative Adversarial Networks) to separate and transfer identity information in the latent space. The underlying generative models, such as StyleGAN [15], produce high-resolution faces with finely controllable attributes. The technical challenge is to create natural transitions between the inserted face and the surrounding context. Particularly at fusion boundaries such as the hairline or jaw, characteristic artifacts emerge that can serve as a basis for detection.

2.1.2. Face Reenactment

Face reenactment transfers the facial movements of one person to the face of another person, while preserving the identity of the target face. This technique makes it possible to attribute to people facial expressions or statements that they have never shown or made. The technical implementation is often based on 3D Morphable Models (3DMMs) that parametrically describe facial geometry. Methods such as Face2Face capture facial movements in real time and transfer them to a target video. Newer approaches combine 3D reconstruction with neural renderers to achieve photorealistic results.

2.1.3. Talking Face Generation

Talking face generation creates speaking people from static images or short video clips, controlled by audio data or text. Models such as Wav2Lip synchronize lip movements with arbitrary audio. This manipulation technique is relevant for the present thesis because it produces both visual and acoustic artifacts. The lip movements are controlled by the audio model, so inconsistencies can arise between the natural speech of the person and the generated mouth movement.

2.1.4. Audio Manipulations

In addition to visual manipulations, the audio tracks of videos can also be altered. Specifically, individual words or sentences are replaced by synthetic speech synthesis, while the visual track remains unchanged. Such manipulations cannot be recognized with the naked eye, since the video image appears authentic. For detection, therefore, analysis of the audio track is necessary. Modern speech synthesis models achieve high quality, but leave characteristic spectral artifacts in the frequency domain.

2.1.5. The Four Manipulation Classes

For multimodal detection, it is relevant which modality of a video was manipulated. Since audio and image tracks can be authentic or fake independently of each other, four classes result:

- **Real (R):** Both audio and visual frames are authentic.
- **Fake Audio, Real Visual (FARV):** Only the audio track is manipulated, the visual track remains authentic.
- **Real Audio, Fake Visual (RAFV):** Only the visual track is manipulated, the audio track remains authentic.
- **Fake Audio, Fake Visual (FAFV):** Both audio and visual frames are manipulated.

This four-class formulation is practically relevant for deepfake forensics, as it not only indicates the presence of manipulation but also names the affected modality. The binary classification can be derived from the four classes at any time by combining all classes except Real as Fake. In this thesis, both formulations are considered. Figure 2.1 illustrates how the four classes result from the manipulation status of both modalities.

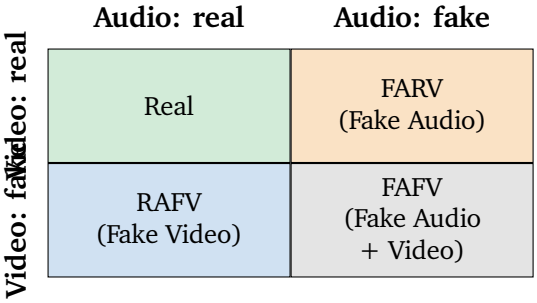


Figure 2.1.: The four classes result from the combination of the manipulation status of audio and video tracks.



Figure 2.2.: Example frames of the four classes (Real, FARV, RAFV, FAFV) from AV-Deepfake1M++ [6].

2.2. Convolutional Neural Networks for Video Analysis

Convolutional Neural Networks (CNNs) are a class of neural networks characterized by local connections and shared weights. Specifically, convolution operations are used to extract spatial features from input data. This makes CNNs suitable for processing image data, since local structures such as edges, textures, and shapes are efficiently captured.

2.2.1. ResNet and Residual Learning

ResNet (Residual Network) introduced residual connections that direct training gradients directly through deep networks and thus solve the vanishing gradient problem. A residual connection adds the input of a block directly to its output:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{2.1}$$

where $\mathbf{x}$ represents the input, $\mathbf{y}$ the output, and $\mathcal{F}$ the residual mapping. ResNet-50 serves in Phase 1 as an encoder for the extraction of frame features. Pre-trained weights from ImageNet enable the use of learned visual representations as a starting point for fine-tuning on deepfake data.

2.2.2. 3D CNNs and X3D

For video analysis, 3D CNNs were developed that process temporal and spatial dimensions together. The 2D convolution is extended to three dimensions, allowing features to be extracted from frame sequences. This enables direct modeling of temporal dependencies without requiring a separate architectural component.

X3D (Expandable 3D Networks) [10] is a family of efficient 3D CNNs developed by gradually expanding 2D networks along various axes. The X3D-M variant offers a balance between computational cost and performance. Compared to other 3D CNNs such as SlowFast or I3D, X3D-M achieves similar recognition rates with lower memory requirements.

2.3. Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks [11] are a special type of recurrent neural network (RNN) that can model long-term dependencies in sequential data. The problem of classical RNNs, the vanishing gradient for long sequences, is solved through so-called gates. An LSTM block consists of an input gate i_t, a forget gate f_t, an output gate o_t, and a cell state c_t:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{2.2}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{2.3}$$
$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{2.4}$$
$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \tag{2.5}$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{2.6}$$
$$h_t = o_t \cdot \tanh(c_t) \tag{2.7}$$

where σ is the sigmoid function and h_{t-1} represents the previous hidden state. The forget gate decides which information from the cell state is discarded, while the input gate adds new information. The bidirectional LSTM processes the frame features from ResNet-50 sequentially in Phase 1. Since deepfake artifacts often appear as temporal inconsistencies between frames, temporal modeling through the LSTM is helpful for detection.

2.4. Transformers and Attention Mechanisms

The Transformer [28] was originally developed for Natural Language Processing and is based exclusively on attention mechanisms without recurrent or convolutional structures. The central component is the Self-Attention mechanism, which directly models relationships between all elements of a sequence.

2.4.1. Self-Attention

The Scaled Dot-Product Attention mechanism computes weights based on the similarity between Query (Q), Key (K), and Value (V) vectors:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2.8}$$

where d_k is the dimension of the key vectors and serves as a scaling factor to keep gradients stable. By directly modeling relationships between all elements, even long-range dependencies can be captured, which with RNNs is only possible step by step.

2.4.2. TimeSformer

TimeSformer [5] extends the Vision Transformer (ViT) to video sequences by applying spatial and temporal attention separately. A video of size $H \times W \times 3 \times F$ is divided into $N = HW/P^2$ patches of size $P \times P$ per frame. The Divided Space-Time Attention first processes temporal attention over all frames at the same spatial position, followed by spatial attention within each frame. This factorization reduces computational cost from $O((NF)^2)$ to $O(N^2 + F^2)$ compared to joint space-time attention.

2.4.3. Cross-Attention for Multimodal Fusion

Cross-attention is an extension of the self-attention mechanism in which query vectors come from one modality and key and value vectors come from another modality. This allows a model to learn which parts of one modality are relevant for the other:

$$\text{CrossAttention}(Q_a, K_v, V_v) = \text{softmax}\left(\frac{Q_a K_v^T}{\sqrt{d_k}}\right) V_v \tag{2.9}$$

where Q_a represents the query vectors of the audio modality and K_v, V_v the key and value vectors of the video modality. Unlike simple concatenation of features from both modalities, cross-attention enables targeted interaction in which one modality conditions the other. This is relevant for detecting audio-visual deepfakes, since inconsistencies between modalities are a detection feature.

2.5. Self-Supervised Learning for Audio

Self-supervised learning (SSL) is a training method in which models are trained without explicitly annotated data by generating pseudolabels directly from the input data. In the audio domain, SSL approaches have made progress in recent years and enable learning of high-quality representations from large amounts of unlabeled audio data.

2.5.1. HuBERT

HuBERT (Hidden-Unit BERT) [12] extends the SSL approach through masked prediction of hidden units. Instead of a contrastive loss, discrete cluster assignments are used as prediction targets, which are generated through k-means clustering in the latent space. The model takes raw waveforms as input and processes them through a convolutional encoder followed by a transformer:

$$\mathcal{L}_m(f; X, M, Z) = \sum_{t \in M} \log p_f(z_t \mid \tilde{X}, t) \tag{2.10}$$

where M denotes the set of masked time steps, z_t the cluster target, and $\tilde{X}$ the masked input. An advantage of this approach lies in the decoupling of acoustic unit discovery from pretraining. This enables an iterative refinement process in which improved cluster assignments are generated in each iteration. HuBERT serves as the audio encoder in all three phases. The model `hubert-base-ls960` pre-trained on 960 hours of

LibriSpeech extracts 768-dimensional feature vectors per time step that encode the acoustic characteristics of the audio signal.

The architecture of the HuBERT model used consists of seven convolutional blocks with temporal convolution layers, layer normalization, and GELU activations, followed by a transformer encoder with 12 layers. Each transformer block processes the convolutional features through multi-head self-attention and a feed-forward layer with residual connections.

2.6. Multimodal Fusion

The fusion of audio and video features is a central step in multimodal detection systems. The separately extracted representations of both modalities are combined into a shared representation. The following describes the fusion strategies used in this thesis.

2.6.1. Feature-Level Concatenation

The simplest fusion strategy is concatenation of the feature vectors of both modalities:

$$\mathbf{f}_{\text{fusion}} = [\mathbf{f}_{\text{video}} \parallel \mathbf{f}_{\text{audio}}] \tag{2.11}$$

where $\mathbf{f}_{\text{video}} \in \mathbb{R}^{d_v}$ and $\mathbf{f}_{\text{audio}} \in \mathbb{R}^{d_a}$ represent the feature vectors of the respective modality. The resulting fusion vector $\mathbf{f}_{\text{fusion}} \in \mathbb{R}^{d_v+d_a}$ is subsequently processed through classification layers. This strategy is simple to implement and stable in training. Phases 1 and 2 use this concatenation.

A disadvantage of simple concatenation is that the model must learn to extract interactions between modalities implicitly from the fused features. There is a risk that a dominant or malfunctioning modality distorts the shared representation.

2.6.2. Cross-Attention Fusion

Cross-attention fusion enables targeted interaction between modalities, in which each modality is conditioned by the other. Here, the video modality is conditioned by audio query vectors and vice versa:

$$\mathbf{f}^*_{\text{video}} = \text{CrossAttention}(Q_v, K_a, V_a) \tag{2.12}$$
$$\mathbf{f}^*_{\text{audio}} = \text{CrossAttention}(Q_a, K_v, V_v) \tag{2.13}$$
$$\mathbf{f}_{\text{fusion}} = [\mathbf{f}^*_{\text{video}} \parallel \mathbf{f}^*_{\text{audio}}] \tag{2.14}$$

Through reciprocal conditioning, the model can learn which audio features are relevant for specific visual regions and vice versa. Phase 3 uses cross-attention to capture these bidirectional dependencies.

2.7. Transfer Learning and Fine-Tuning

Transfer learning refers to the transfer of knowledge that a model has learned on a source task to a related target task. Here, the weights of a pre-trained model are used as initialization instead of training the model from scratch. This reduces the need for labeled training data and shortens training time.

In this thesis, all models used are initialized with pre-trained weights. ResNet-50 is initialized with weights from ImageNet pre-training, X3D-M and TimeSformer use weights pre-trained on Kinetics-400, and HuBERT starts with weights from pre-training on 960 hours of LibriSpeech. This pre-training on large datasets enables the models to learn general visual or acoustic representations that are subsequently adapted to deepfake detection.

During fine-tuning, not the entire model is trained simultaneously. Instead, only the upper layers are released initially, while deeper layers remain frozen. This prevents the learned general representations from being overwritten by the deepfake-specific training process. This conservative strategy improves generalization to unknown manipulation techniques.

2.8. Evaluation Metrics

Three metrics are used to assess detection performance in this thesis. Accuracy (ACC) indicates the proportion of correctly classified examples. However, for unbalanced datasets, accuracy is not very meaningful, since a model can achieve high values simply by predicting the majority class.

Unweighted Average Recall (UAR) averages recall over all classes unweighted and is therefore more robust to class imbalances:

$$\text{UAR} = \frac{1}{C} \sum_{c=1}^{C} \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{2.15}$$

where C denotes the number of classes, TP_c the correctly and FN_c the incorrectly undetected examples of class c. A model that predicts only the majority class achieves a UAR of only the chance level.

The Area Under the ROC Curve (AUC) measures a model's ability to distinguish between classes independent of a fixed decision threshold. A value of 0.5 corresponds to random classification, while a value of 1.0 indicates perfect separation. AUC is particularly suitable for comparing models because it considers the full range of possible thresholds. In the four-class formulation, AUC is averaged via a one-vs-rest approach.

3. Related Work

The detection of deepfakes has gained importance in recent years and is summarized in several overview works [29]. This chapter analyzes existing approaches. The presentation is divided into four areas: datasets for deepfake detection, unimodal detection methods, multimodal approaches for video-level classification, and approaches for temporal localization, which are presented as a contrasting research direction. At the end, the research gap addressed by the present thesis is identified.

3.1. Datasets for Deepfake Detection

The development of effective detection methods requires extensive training data. Early datasets such as DF-TIMIT and UADFV contained only a few hundred to several thousand videos and focused primarily on face-swapping manipulations [29]. FaceForensics++ [27] extended this approach and was the first to cover various manipulation techniques including Face2Face, FaceSwap, and NeuralTextures. The dataset became established as a standard benchmark for evaluating detection methods.

The DFDC Dataset [9] was among the first large-scale datasets in deepfake research. With over 100,000 videos and 3,426 speakers, the data volume enabled training of more complex models. Datasets such as Celeb-DF [21] and DeeperForensics-1.0 [13] followed with the goal of providing more realistic manipulations and higher quality.

A characteristic of these early datasets is that they are designed exclusively for video-level classification and only manipulate the visual modality. FakeAVCeleb [16] extended the focus to multimodal manipulations by combining face-swapping and face-reenactment with audio manipulations. The dataset builds on real videos from VoxCeleb2 [25] and includes 500 authentic and over 19,500 forged videos of celebrities. Since authentic videos are greatly underrepresented compared to manipulated ones, FakeAVCeleb is heavily imbalanced, which must be taken into account when interpreting results.

The DeepSpeak Dataset [4] focuses on high-quality webcam recordings of people speaking and gesturing. Version 2.0 includes 280 people and various deepfake types, including avatar generation, face-swapping, lip-sync, and audio deepfakes. The dataset is natively three-class in structure: Since audio deepfakes were inserted only into a subset of lip-sync deepfakes [4], manipulated audio occurs only in combination with manipulated video. The FARV class, namely manipulated audio track with unchanged video, is thus not present, so a full four-class formulation is not possible. For this reason, DeepSpeak is used in this thesis exclusively for binary classification.

The AV-Deepfake1M++ Dataset [6] represents the state of the art. With over 2 million videos and more than 2,000 speakers, it is the largest available dataset for audio-visual deepfake detection. It includes three manipulation strategies, replacement, deletion, and insertion, and contains all four classes Real, FARV, RAFV, and FAFV. This makes it suitable for both binary and four-class classification and forms the primary training

dataset for this thesis. Evaluation on this dataset shows that existing methods achieve worse results than on earlier datasets, which is attributable to the higher quality of manipulations and the variability of segment lengths.

Table 3.1.: Comparison of important deepfake detection datasets

Dataset	Year	Videos	Subjects	Modality
FaceForensics++	2019	5,000	–	Video
DFDC	2020	128,154	3,426	A+V
Celeb-DF	2020	6,229	59	Video
DeeperForensics	2020	60,000	100	Video
FakeAVCeleb	2021	20,000+	600+	A+V
DeepSpeak v2.0	2025	16,585	280	A+V
AV-Deepfake1M++	**2024**	**2M+**	**2,000+**	**A+V**

3.2. Unimodal Detection Methods

Early detection methods were based on the analysis of artifacts in individual images. MesoNet [1] used a compact CNN architecture with only four layers to detect face-swapping manipulations. The method analyzes inconsistencies at the mesoscopic level, such as unnatural textures at fusion boundaries. Despite the small model size, MesoNet achieved good results on early datasets, but showed weaknesses with higher-quality manipulations.

Blending artifacts, which arise from the fusion of different facial regions, are the focus of Face X-ray [20]. The method generates a visual representation of fusion boundaries and uses it as an auxiliary signal for classification. Since blending artifacts occur regardless of the specific generation method, various face-swapping techniques can be detected without the method being tailored to a single technique.

As generation technology improved, pixel-based artifacts became more difficult to detect. Zhao et al. [33] take a different approach: Instead of a single classifier, multiple attention maps are computed in parallel, each focusing on different facial regions and capturing both spatial and temporal features. Zheng et al. [34] investigated temporal coherence in deepfake videos and used 3D CNNs to analyze frame sequences in order to detect unnatural movement patterns.

Another source of artifacts lies in the frequency domain. Li et al. [19] proposed a method for extracting frequency-based features to detect compression artifacts and periodic patterns resulting from GAN-based generation. Liu et al. [23] extended this approach by combining spatial and phase features in the frequency domain.

Another group of procedures uses biological or identity-related signals. DeepVision [14] analyzes blinking behavior, since synthesized faces often exhibit unnatural blinking patterns. TI2Net [22] models temporal identity inconsistencies across the video sequence.

These unimodal methods achieve good results on their respective benchmark datasets, but show limitations in generalization to new manipulation techniques and fundamentally cannot detect pure audio manipulations since they do not consider the audio track. Additionally, CNN-based detectors are vulnerable to adversarial perturbations [24].

3.3. Multimodal Video-Level Detection

Multimodal approaches use both audio and video to enable holistic analysis. They form the relevant category for this thesis, as classification at the video level is performed here while incorporating both modalities.

An early example of joint consideration of both modalities comes from Agarwal et al. [2] on phoneme-viseme inconsistencies. Phonemes are the smallest units of spoken language, while visemes are the corresponding visible mouth movements. The method analyzes whether lip movements match the spoken sounds and thereby detects talking face generation, which produces characteristic asynchronies.

Cheng et al. [8] investigated voice-face homogeneity, namely the natural match between a person's voice and face. The method extracts speaker embeddings from audio and facial features from video and analyzes the correlation between the two modalities. This is effective for face-swapping detection because when identities are swapped, the voice does not match the visual identity.

Zou et al. [35] developed cross-modality and within-modality regularization (MRDF) for audio-visual deepfake detection. The cross-modality module minimizes the distance between paired audio-video features of authentic videos and maximizes it for manipulated pairs. Evaluation on FakeAVCeleb showed that particularly the detection of RAFV videos is improved by explicit modeling of modality relationships, a class in which purely visual methods fail.

The AV-LGNN system [30] combines local and global interactions between audio and video. Here, video Swin Transformer and EResNet are used as encoders. The system first performs local interaction within regions and subsequently global interaction between all regions, with cross-attention applied in the final step. Additionally, an adaptive modality selection is proposed that dynamically selects the most informative modality for each video.

With HOLA, Wu et al. [31] introduced a system that combines hierarchical contextual aggregations with self-supervised pre-training. The iterative-aware cross-modal learning module performs multiple successive fusion steps, with each step refining the representations of both modalities. Evaluation on AV-Deepfake1M++ showed that HOLA achieves better results than existing methods on this dataset.

3.4. Approaches to Temporal Localization

Part of current research does not pursue video-level classification but temporal localization of manipulations. It is determined not only whether a video is manipulated but also in which time intervals the manipulation occurs. This research direction is presented below as a contrast, since the present thesis deliberately focuses on video-level classification and does not pursue temporal localization.

Cai et al. [7] developed BA-TFD (Boundary Aware Temporal Forgery Detection) as the first specialized approach for localizing content-driven manipulations on the LAV-DF dataset. The method combines contrastive learning, frame-level classification, and a boundary matching network. BA-TFD achieved 96.30% AP@0.5 on LAV-DF, but fell to 14.70% AP@0.5 when switching to AV-Deepfake1M++, indicating generalization problems.

Zhang et al. [32] proposed UMMAFormer, a universal multimodal-adaptive transformer for temporal forgery localization. The method uses parallel cross-attention feature pyramid networks to process features at multiple temporal resolutions. UMMAFormer achieved 98.83% AP@0.5 on LAV-DF but showed a drop to 51.64%

AP@0.5 on AV-Deepfake1M++. This difference illustrates that localization methods optimized on one dataset are not readily transferable to larger and more diverse datasets.

The Vigo system [26] combines audio and video for segment-level localization. For audio analysis, Vigo uses Wav2Vec2.0 [3] and a gMLP-based backend; for video analysis, temporal segment networks. Vigo achieved good results in the ACM MM 1M deepfakes detection challenge for the localization task and showed that an efficiency-focused architecture generalizes better on AV-Deepfake1M++ than more complex transformer approaches.

A deliberately lean approach is taken by KLASSify by Kukanov and Ng [18]: SSL-based audio features from Wav2Vec-AASIST are combined with hand-crafted visual features processed through temporal convolutional networks. Evaluation on AV-Deepfake1M++ showed 92.78% AUC in video-level classification.

Klein et al. [17] proposed Pindrop, a system for fine-grained localization of manipulations. The approach processes audio and video in parallel and uses multi-task learning with separate heads for detection and temporal localization. This way, the video-level decision and the determination of manipulated sections are learned in a single model.

3.5. Research Gap

The analysis of existing work reveals several limitations that the present thesis addresses.

First, there is a lack of systematic comparison of different video encoders and fusion strategies under identical conditions. The presented multimodal systems differ in architecture, training data, and evaluation protocol, making it difficult to isolate the contribution of individual components. A comparison that varies only the video encoder and fusion strategy while keeping the audio branch and training procedure constant is rare in the literature.

Second, cross-dataset generalization is usually treated as secondary. The cited works report primarily in-domain results. Existing generalization studies in the field of temporal localization [6] indicate pronounced performance drops, but a systematic investigation across all dataset combinations and separately by modality is lacking.

Third, the different contribution of audio and video modality to generalization is rarely considered separately. Since fusion combines both modalities, it remains unclear which modality carries generalization and which impairs it.

The present thesis addresses these gaps through a systematic comparison of three architectures, from ResNet-50 with LSTM to X3D-M to TimeSformer, with constant audio branch and training procedure. All models are evaluated in-domain and across all dataset combinations cross-dataset, with performance additionally evaluated separately by modality.

4. Design

This chapter describes the developed detection system. First, the overall structure shared by all three phases is presented. Subsequently, the preprocessing pipeline, the shared audio branch, and the three phases are explained in detail. For each phase, the architecture is explained and the associated design decisions are justified. The description also includes the preprocessing pipeline, feature extraction through video and audio encoders, and fusion strategies.

4.1. Overall Structure

The system processes a video as a pair of an image sequence and the associated audio track. Both modalities are processed through separate encoders, whose outputs are fused into a shared representation. Based on this representation, a classification head makes the final decision. The classification is at the video level, meaning a single class assignment is output for the entire video. Both binary classification and four-class classification use this structure, differing only in the number of output neurons and label assignment.

The three investigated architectures are referred to hereafter as phases and can be summarized as follows. Phase 1 extracts features from individual frames with a ResNet-50, models their temporal progression with a bidirectional LSTM, and fuses video and audio features through concatenation. Phase 2 replaces this two-stage video branch with the 3D-CNN X3D-M, which processes spatial and temporal information jointly, while retaining concatenation. Phase 3 uses TimeSformer as a transformer-based video encoder and additionally replaces concatenation with cross-attention fusion.

The fact that multiple phases investigate only the video branch is a deliberate decision. On one hand, the space of possible video encoders is diverse: frame-based CNNs with separate temporal modeling, 3D-CNNs, and video transformers represent fundamentally different processing principles, whose influence on detection performance is one of the central questions of this work. On the other hand, for audio processing with self-supervised pre-trained speech models, an established approach already exists that is consistently employed in related work. The audio component therefore does not have multiple phases but instead consistently uses a predefined self-supervised learning model—in this thesis, HuBERT—which would have been interchangeable with comparable models such as wav2vec 2.0 [3] or WavLM. A constant audio branch also provides a methodological advantage: performance differences between phases can be uniquely attributed to the video encoder and fusion strategy.

The three phases accordingly share the audio branch, training procedure, and preprocessing pipeline. They differ exclusively in the video encoder and, in the case of Phase 3, in the fusion strategy.

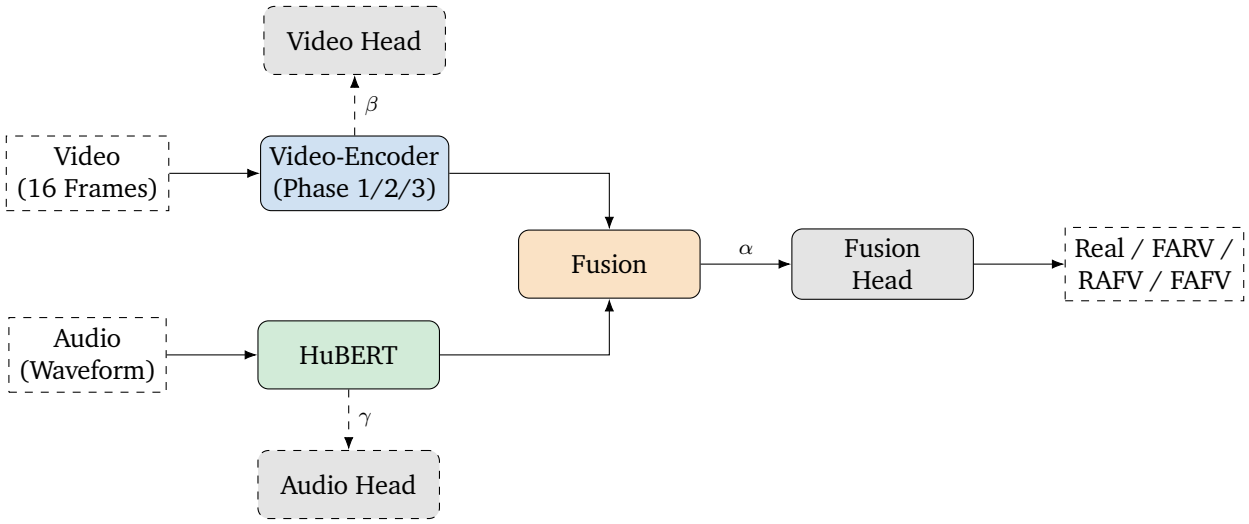


Figure 4.1.: Overall structure of the detection system. The video encoder is exchangeable between phases, while the audio branch (HuBERT) and the classification head remain constant.

4.2. Preprocessing and Data Caching

From each video, the visual and acoustic modalities are extracted separately. For the visual modality, 16 frames are sampled over the video length and scaled to uniform resolution. Uniform sampling is used as the standard, distributing frames evenly across the video. Even distribution ensures that the sampled frames cover the entire temporal extent of the video, even if manipulation affects only part of it. For the acoustic modality, the audio signal is resampled to 16 kHz and brought to a fixed length of 64,000 samples, corresponding to four seconds.

As an alternative, segment-based sampling was implemented that deliberately samples a large proportion of frames from the known manipulation segments for manipulated videos. This variant assumes temporal manipulation boundaries are known and serves as a comparison point in Chapter 5. For actual video-level detection, where these boundaries are not available at inference time, uniform sampling is used.



Figure 4.2.: Comparison of sampling strategies on a manipulated video. Top shows uniformly sampled frames; bottom shows segment-based sampled frames. The manipulated segment is highlighted.

Since the three datasets combined have large volume and feature extraction through the pre-trained encoders is computationally intensive, preprocessed inputs are stored in a sharded HDF5 cache. Data is divided into multiple shards allowing efficient parallel loading. This cache is used for all datasets and ensures preprocessing and training are decoupled. This reduces training time and guarantees all phases train on identically preprocessed data.

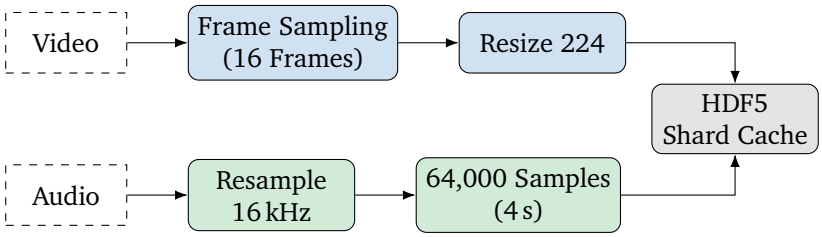


Figure 4.3.: Preprocessing pipeline for video and audio modality with subsequent HDF5 shard caching.

4.3. Audio Branch

The audio branch is identical across all three phases and is based on HuBERT [12]. The model `hubert-base-ls960`, pre-trained on 960 hours of LibriSpeech, processes the raw waveform and produces a sequence of 768-dimensional feature vectors. This sequence is aggregated into a fixed representation that is subsequently passed to fusion.

HuBERT was chosen because the model provides robust and general speech representations through prediction of discrete cluster targets (see Section 2.5) and has proven effective in related work on audio deepfake detection. Comparable self-supervised models such as wav2vec 2.0 [3], used for example in the Vigo system [26], or WavLM would have been alternatives and could have been substituted without changing the rest of the architecture due to identical output dimension of 768. Since the audio branch is kept constant across all phases to attribute performance differences to the video branch and fusion, comparison of multiple audio encoders was foregone.

Self-supervised pre-trained models on large speech corpora such as HuBERT provide representations not tailored to a specific deepfake dataset but encoding general acoustic properties of speech. This property is relevant for generalization because it enables the audio branch to work on datasets not seen during training. Fine-tuning releases only the last four of twelve transformer layers while lower layers remain frozen. This preserves general representations learned in pre-training.

4.4. Phase 1: ResNet-50 with BiLSTM

Phase 1 processes the visual modality in two steps. First, a ResNet-50 extracts a feature vector from each of the 16 frames. Since ResNet-50 is a 2D-CNN, frames are processed individually without considering temporal information. Subsequently, a bidirectional LSTM processes the sequence of frame features and thereby models temporal dependencies between frames. Bidirectionality allows each frame to be considered in the context of both preceding and following frames.

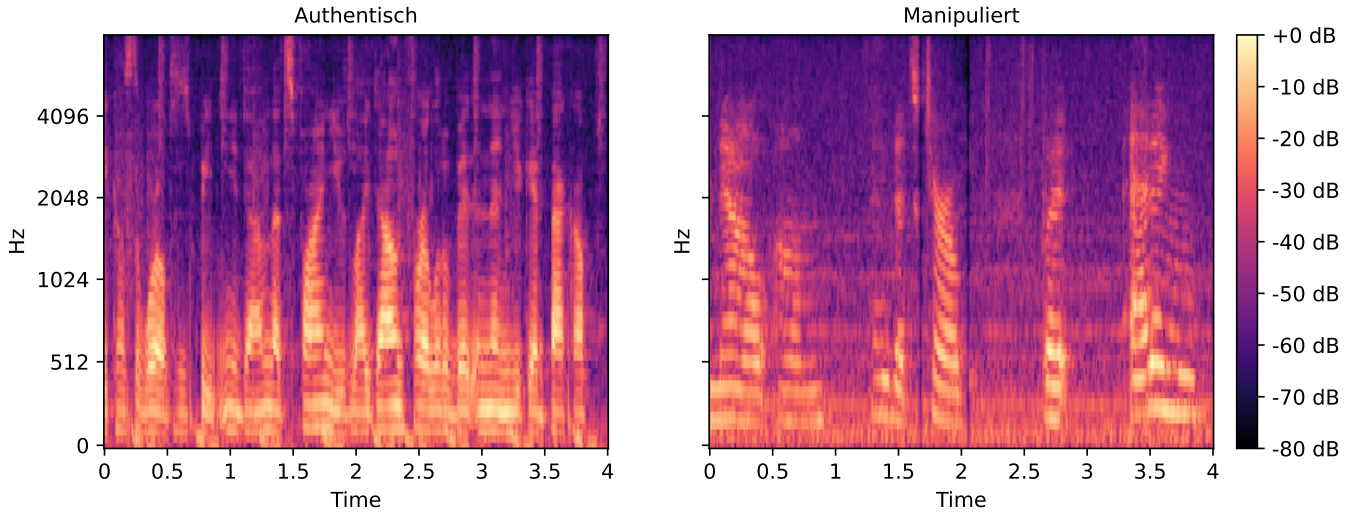


Figure 4.4.: Mel-spectrogram of authentic (left) and synthesized (right) audio signals.

This separation of spatial and temporal modeling is the simplest of the three approaches investigated and serves as a starting point. The motivation is that deepfake artifacts often appear as temporal inconsistencies between successive frames, such as flickering at fusion boundaries or unnatural movement transitions. The BiLSTM is suited to capture such inconsistencies. The disadvantage of this approach is that spatial and temporal modeling occur separately and ResNet-50 has no access to temporal information.

The choice of individual components follows the goal of providing Phase 1 as a simple and well-understood reference architecture. ResNet-50 was chosen as frame encoder because it is an established standard model with freely available ImageNet weights and serves as backbone in numerous detection works. Alternatives such as EfficientNet or deeper ResNet variants would have increased parameter count without providing advantage for the role as reference. For temporal modeling, a bidirectional LSTM was preferred over a GRU since separate cell state allows finer control of information transported across sequence; a transformer encoder as temporal component would have already anticipated the approach of Phase 3. Concatenation was chosen as fusion strategy because it is the simplest and most stable in training and serves as reference against which cross-attention fusion in Phase 3 is compared. Fusion at logit level was rejected in preliminary experiments as it proved less stable (see Section 4.8).

ResNet-50 provides 2048-dimensional vectors per frame. The subsequent bidirectional LSTM has hidden size of 256 and produces 512-dimensional video feature vectors through concatenation of both directions. This is combined with the 768-dimensional audio feature vector through concatenation to a 1280-dimensional representation subsequently processed by a classification head. Only upper blocks of pre-trained ResNet-50 are released while lower layers remain frozen.

4.5. Phase 2: X3D-M

Phase 2 replaces the combination of ResNet-50 and BiLSTM with the 3D-CNN X3D-M [10]. Unlike Phase 1, X3D-M processes spatial and temporal dimensions together. Convolution operations extend across multiple

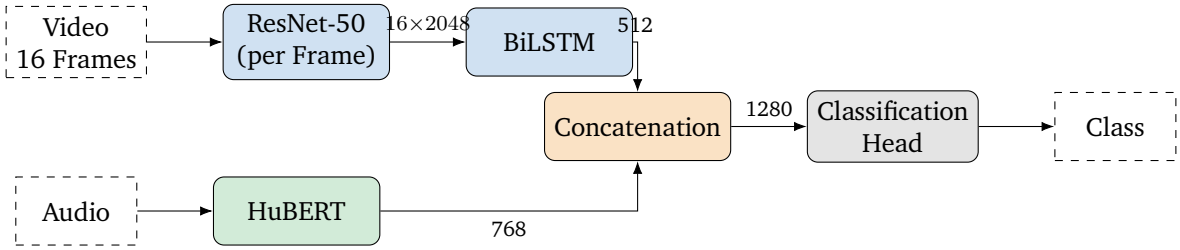


Figure 4.5.: Architecture of Phase 1. ResNet-50 extracts frame features, a BiLSTM models the temporal dimension, HuBERT processes audio. Fusion is through concatenation.

frames so the network extracts motion information directly from the video sequence without requiring a separate temporal component like LSTM.

The switch to X3D-M follows from a weakness of Phase 1: Separate processing limits interaction between spatial and temporal features. A 3D-CNN can capture spatio-temporal patterns such as inconsistent mouth movements that only emerge from joint consideration of multiple frames. X3D-M was chosen because among 3D-CNNs it offers balance between recognition performance and memory requirements, which is relevant given limited GPU resources and dataset sizes. Alternatives such as SlowFast or I3D achieve comparable recognition rates on standard benchmarks but require more memory and compute time [10]. Since each phase required multiple training runs across three datasets and two task formulations, X3D-M's efficiency was decisive. The audio branch and concatenation fusion remain unchanged from Phase 1, so the difference between both phases lies exclusively in the video encoder.

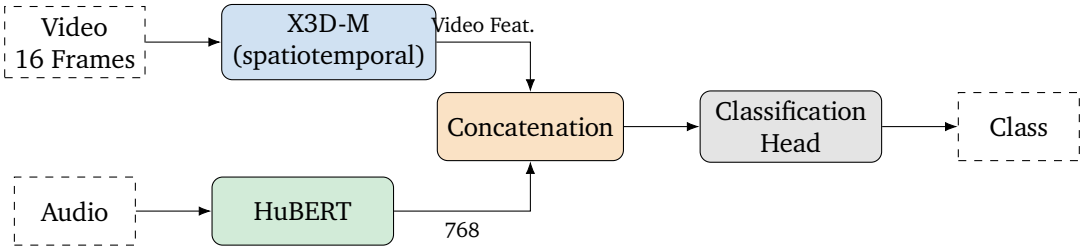


Figure 4.6.: Architecture of Phase 2. X3D-M processes the video sequence spatiotemporally, HuBERT processes audio. Fusion is through concatenation.

4.6. Phase 3: TimeSformer with Cross-Attention

Phase 3 uses TimeSformer [5] as video encoder and additionally replaces concatenation with cross-attention fusion. TimeSformer applies divided space-time attention described in Section 2.4 and models long-range temporal dependencies across all frames. Unlike local receptive fields of CNNs, the attention mechanism can directly capture relationships between distant frames.

TimeSformer was chosen among video transformers because divided space-time attention reduces computational cost compared to joint space-time attention and pre-trained weights on Kinetics-400 are freely available. Alternatives such as ViViT or Video Swin Transformer, used for example in AV-LGNN [30], pursue related concepts; however, TimeSformer offered best availability of pre-trained models in multiple input resolutions, enabling the comparison of resolutions 224 and 448 described in Section 5.6.

The second change concerns fusion. Instead of concatenating feature vectors of both modalities, cross-attention fusion conditions each modality on the other. For this, both modalities are first projected to a shared dimension of 512. Subsequently, reciprocal multi-head attention operations are performed in two layers with eight attention heads each, where video representation uses audio representation as keys and values and vice versa. Each attention operation is accompanied by a residual connection and layer normalization. The resulting representations are averaged and concatenated. Behind cross-attention lies the consideration that audio-visual deepfakes often exhibit inconsistencies between modalities, such as between lip movement and audio signal. A fusion that explicitly relates both modalities to each other is better suited to model such inconsistencies than simple concatenation, which leaves interaction to the classification head. Alternative strategies such as gated fusion, which weights modalities depending on reliability, or bilinear pooling were not implemented since the comparison aimed to directly contrast cross-attention with simple concatenation. Figure 4.7 shows the structure of cross-attention fusion.

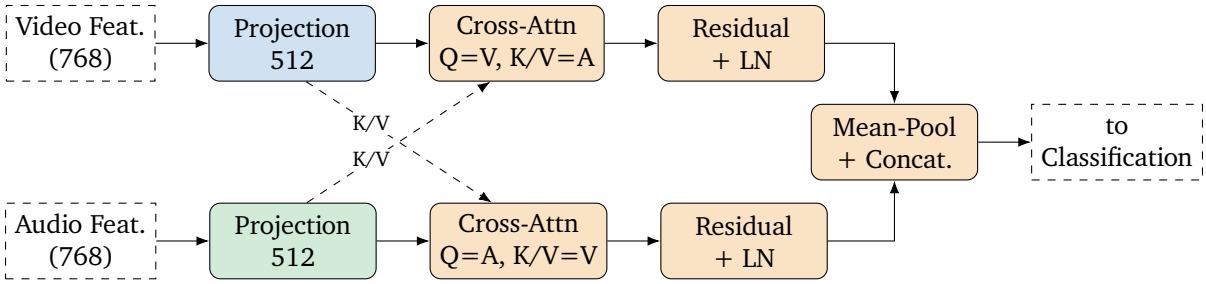


Figure 4.7.: Structure of cross-attention fusion. Both modalities are projected to shared dimension and reciprocally conditioned on each other before representations are averaged and concatenated.

Phase 3 is thus the most complex of the three architectures. It combines the most expressive video encoder with the most elaborate fusion strategy. Whether this additional complexity leads to better recognition performance is investigated in Chapter 5.

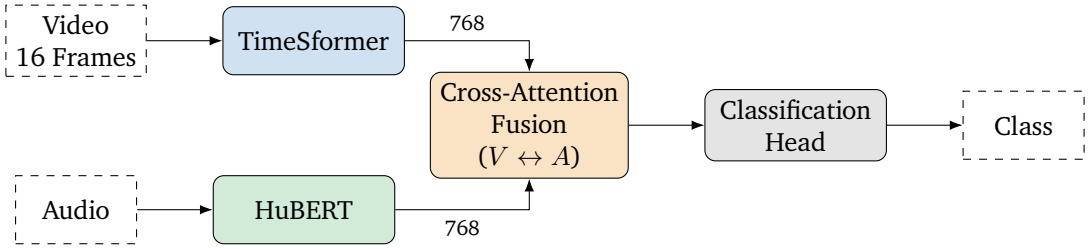


Figure 4.8.: Architecture of Phase 3. TimeSformer processes the video sequence, HuBERT processes audio, and cross-attention fusion reciprocally conditions both modalities on each other.

4.7. Auxiliary Losses and Classification

To ensure that both modalities learn meaningful representations, in addition to loss on the fused representation, an additional loss is computed on both video and audio branches. These auxiliary losses force each branch to also make classifications without the other modality. Total loss consists of a weighted sum:

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{fusion}} + \beta\,\mathcal{L}_{\text{video}} + \gamma\,\mathcal{L}_{\text{audio}} \tag{4.1}$$

with weights $\alpha = 0.6$, $\beta = 0.2$, and $\gamma = 0.2$. Fusion loss receives the highest weight since it represents the actual target, while the two modality-specific losses serve as regularization. Without these auxiliary losses, there is risk that the model relies on a dominant modality and neglects the other. The modality-separated evaluation in Chapter 5 uses the outputs of the two additional classification heads to quantify the contribution of each modality individually.

For four-class classification, each class label results from the combination of both modality states. Manipulation of the visual modality occurs in classes RAFV and FAFV, manipulation of the audio track in classes FARV and FAFV. From this assignment, the targets of the two modality-specific classification heads can be derived so the video head trains on visual manipulation and the audio head on acoustic manipulation.

4.8. Design Decisions

In addition to the architecture, several training and design decisions were made that affect stability and generalization. The following decisions are based on preliminary experiments conducted in the course of this work.

Frozen Batch Normalization. Even when deeper layers of video encoders are released for fine-tuning, batch normalization layers remain frozen with their running statistics learned in pre-training. The statistics of deepfake data differ from pre-training data. In preliminary experiments, updating running statistics on comparatively small batches led to unstable training with strongly fluctuating validation metrics. Freezing batch normalization stabilized fine-tuning without affecting in-domain performance.

Conservative Unfreezing. Only upper layers of pre-trained video encoders are released, while lower layers remain frozen. This conservative strategy prevents general visual representations from being overwritten by training on a single deepfake dataset. In preliminary experiments, fully unfreezing all layers led to stronger adaptation to training data: while training metrics continued rising, validation UAR stagnated or declined, indicating overfitting. The conservative variant was therefore retained for all phases.

Feature-Level Fusion with Auxiliary Losses. Fusion occurs at feature level rather than logit level. In combination with auxiliary losses, this forces both modalities to produce meaningful representations before fusion. This construction proved more stable in preliminary experiments than pure logit-level fusion, where modalities were combined only at the end. With logit fusion, the model repeatedly relied on the in-domain dominant modality while the other branch learned no meaningful representations.

Pre-trained Initialization. All encoders are initialized with pre-trained weights. Given the difficulty of the task and limited training data per dataset, training from scratch is not practical. Pre-training provides a starting representation adapted to deepfake detection through fine-tuning.

5. Evaluation

This chapter presents the experimental results. First, the evaluation setup is described. Subsequently, in-domain results per phase, cross-dataset generalization, and modality-separated analysis are presented. Finally, the results are interpreted.

5.1. Setup

5.1.1. Datasets

Evaluation is conducted on three datasets. AV-Deepfake1M++ [6] serves as primary training dataset and is used in both binary and four-class formulations. Approximately 50,000 videos are used for training and approximately 40,000 videos for validation from the dataset, with the four classes approximately equally distributed. FakeAVCeleb is trained in four-class formulation and is heavily imbalanced by the low proportion of authentic videos. DeepSpeak v2.0 [4] is trained exclusively in binary form since the dataset is natively three-class and does not provide the FARV class: manipulated audio occurs there only in combination with manipulated video, not with unchanged video material.

5.1.2. Implementation and Training

Implementation is based on PyTorch version 2.5.1 with CUDA 12.1 under Python 3.10, the HuggingFace Transformers library for HuBERT, and torchaudio for audio processing. AdamW optimizer with cosine-warmup scheduler is used. To reduce memory requirements, automatic mixed precision (AMP) and gradient accumulation over two steps are employed. All experiments were conducted on a Linux server under Ubuntu 22.04 with a single NVIDIA H100 with 80 GB memory; as processor, two Intel Xeon Platinum 8480+ with approximately 4 TB RAM served. Total loss comprises weighted fusion, video, and audio losses according to Equation 4.1.

5.1.3. Metrics

Accuracy (ACC), Unweighted Average Recall (UAR), and Area Under the ROC Curve (AUC) are used as metrics, as described in Section 2.8. This combination is tailored to task properties: Accuracy serves as intuitive reference but is biased for unbalanced datasets, making UAR necessary as class-balanced metric, particularly for FakeAVCeleb and DeepSpeak. AUC complements both metrics since it works independent of decision threshold and makes models comparable across datasets with different class distributions. All models are evaluated separately by video branch, audio branch, and fusion to assess each modality's contribution

individually. For cross-dataset evaluation, only AUC is reported since it is threshold-independent. A decision threshold calibrated on one dataset is not transferable to others, so accuracy and UAR would be biased under transfer.

5.1.4. Computational Cost

The three phases differ in parameter count. Table 5.1 lists total parameters, trainable parameters, and training time for a run of 20 epochs on AV-Deepfake1M++ for each.

Table 5.1.: Computational cost of the three phases. Training time for one binary run of 20 epochs on an NVIDIA H100 with 80 GB, at resolution 224 pixels unless otherwise noted.

Phase	Video Encoder	Parameters	Trainable	Training Time
Phase 1	ResNet-50 + BiLSTM	124.1 M	57.8 M	∼16.0 h
Phase 2	X3D-M	97.5 M	31.4 M	∼14.6 h
Phase 3	TimeSformer + Cross-Attn.	217.3 M	104.5 M	∼14.5 h
Phase 3 (448)	TimeSformer (HR)	217.3 M	104.5 M	∼54.7 h

Noteworthy is that Phase 3 with 217 million parameters is nearly double Phase 1 and more than double Phase 2. At resolution 224 pixels, training times are nonetheless close together since memory requirements are limited through gradient checkpointing and automatic mixed precision. An exception is the variant at resolution 448 pixels, whose training required approximately 55 hours, nearly four times as much, since image tokens grow quadratically with resolution. The increased parameter count of Phase 3 leads, as shown in following sections, not to better detection performance.

5.1.5. Training Progress

Figure 5.1 and Figure 5.2 show validation UAR progression over epochs for the three phases on AV-Deepfake1M++ and FakeAVCeleb. The progression illustrates that the best epoch is not necessarily the last. In multiple runs, validation metric reaches maximum before training completion, after which slight decline occurs. For evaluation, the model with highest validation UAR is used.

5.2. In-Domain Results

Table 5.2 shows in-domain performance of fusion for all three phases. Each model is evaluated on the dataset it was trained on.

On AV-Deepfake1M++, Phase 1 achieves AUC of 0.968 in binary formulation. This value confirms the pipeline functions and models solve the task fundamentally. Phase 2 is slightly lower at 0.948. Phase 3 drops to AUC of 0.769. Notably, the most elaborate of the three variants performs worst in-domain.

In four-class formulation, performance on AV-Deepfake1M++ expectedly decreases since the task is harder. AUC values range between 0.709 and 0.824, with phase ranking preserved. The comparatively low accuracy

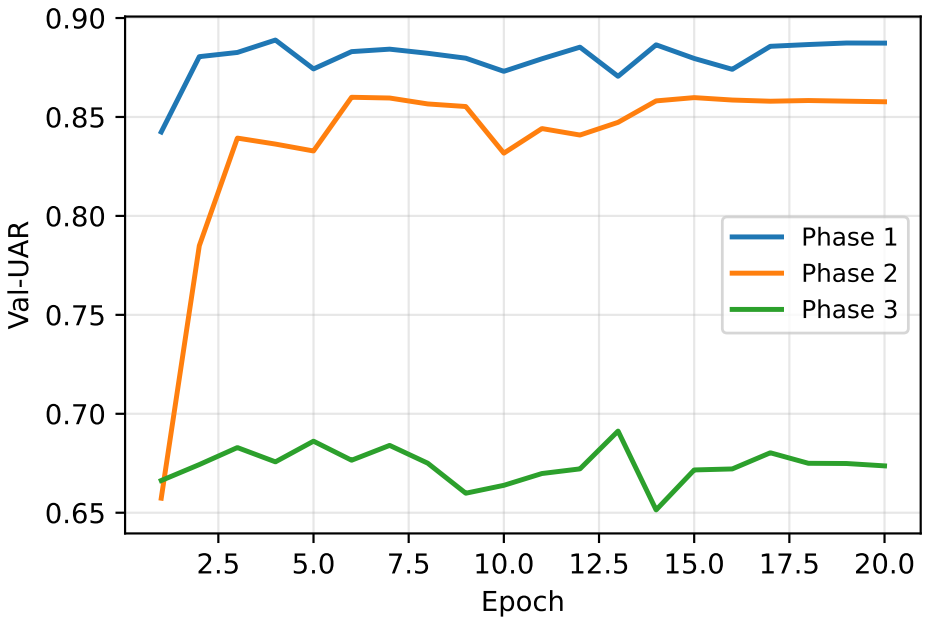


Figure 5.1.: Validation UAR of the three phases over epochs on AV-Deepfake1M++.

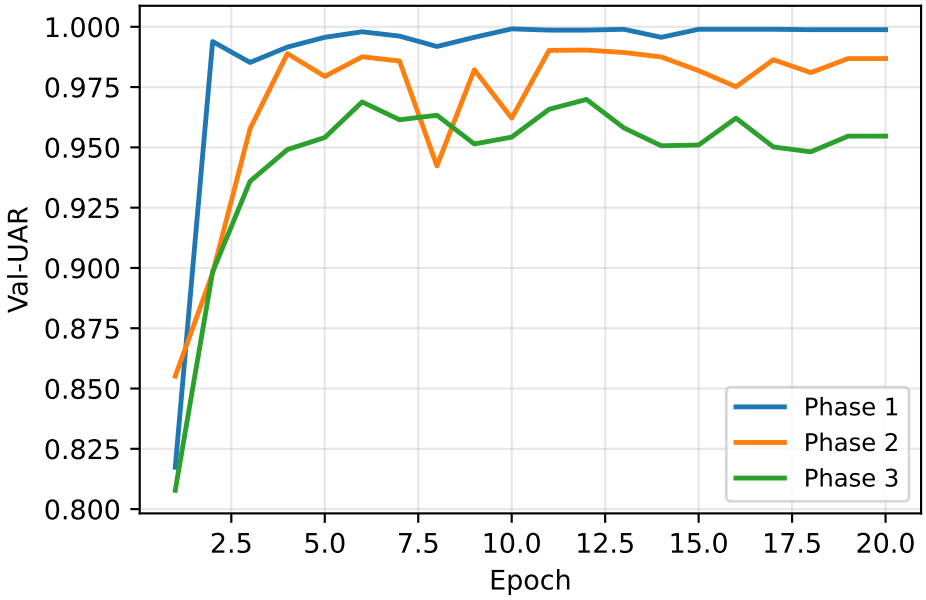


Figure 5.2.: Validation UAR of the three phases over epochs on FakeAVCeleb.

Table 5.2.: In-domain results of fusion per phase. All values on validation split of respective training dataset.

Dataset / Task	Phase	ACC	UAR	AUC
AV-Deepfake1M++ (binary)	Phase 1	88.9%	88.9%	0.968
	Phase 2	86.0%	86.0%	0.948
	Phase 3	69.1%	69.1%	0.769
AV-Deepfake1M++ (4-class)	Phase 1	54.9%	54.9%	0.824
	Phase 2	52.2%	52.2%	0.810
	Phase 3	40.2%	40.2%	0.709
FakeAVCeleb (4-class)	Phase 1	99.8%	99.9%	0.9998
	Phase 2	99.4%	99.0%	0.9998
	Phase 3	98.5%	97.0%	0.994
DeepSpeak v2.0 (binary)	Phase 1	92.6%	93.0%	0.980
	Phase 2	92.4%	93.1%	0.967
	Phase 3	92.7%	93.2%	0.979

around 50% illustrates that particularly distinguishing between both audio-manipulated classes, FARV and FAFV, is difficult.

On FakeAVCeleb, all three phases achieve AUC values near 1.0. This result must be interpreted in context of strong class imbalance. Since authentic videos are rare in FakeAVCeleb and manipulations visually stand out clearly, the task on this dataset is comparatively easy. The high values are therefore less informative than results on AV-Deepfake1M++.

On DeepSpeak, all three phases reliably learn the task. Fusion AUC is 0.980 for Phase 1, 0.967 for Phase 2, and 0.979 for Phase 3, with UAR values around 93%. In-domain, DeepSpeak thus presents no problem for any architecture. The difficulty of this dataset manifests in cross-dataset evaluation in Section 5.4.

5.3. Comparison with Related Work

Direct comparison with published results is only partially possible since used data splits, training data amounts, and sometimes task formulation differ. The following contextualization serves therefore not to demonstrate new state-of-the-art but to provide context. It occurs separately for the three datasets.

5.3.1. AV-Deepfake1M++

The official competition for AV-Deepfake1M++ uses AUC of binary video-level detection on TestA split. Table 5.3 compares values achieved in this work with published results.

Note that published methods trained on full training set with approximately 1.1 million videos while this work used subset of approximately 50,000 videos. Additionally, values in the present work refer to validation split while competition results refer to TestA split. Under these limitations, Phase 1 with AUC of 0.968 lies

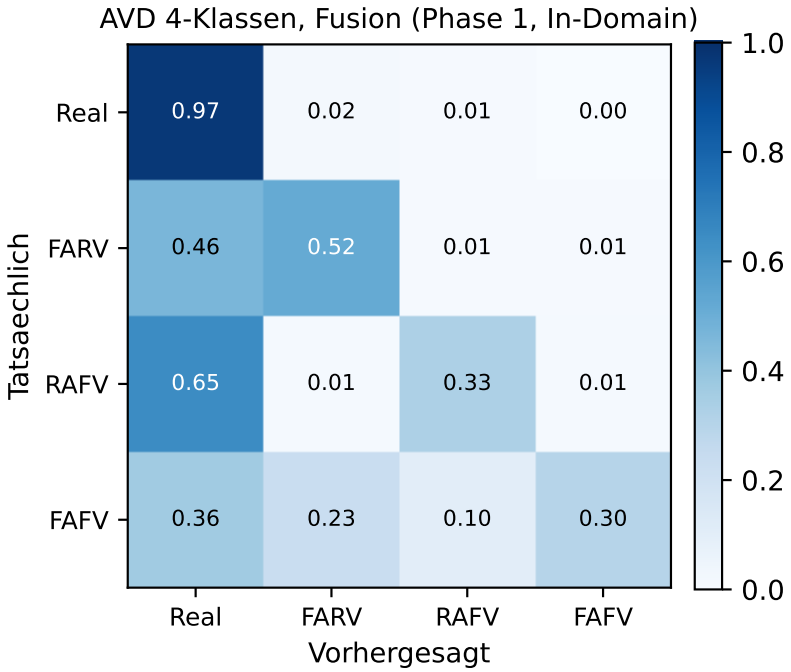


Figure 5.3.: Normalized confusion matrix of four-class fusion of Phase 1 on AV-Deepfake1M++. Confusion between audio-manipulated classes FARV and FAFV is apparent.

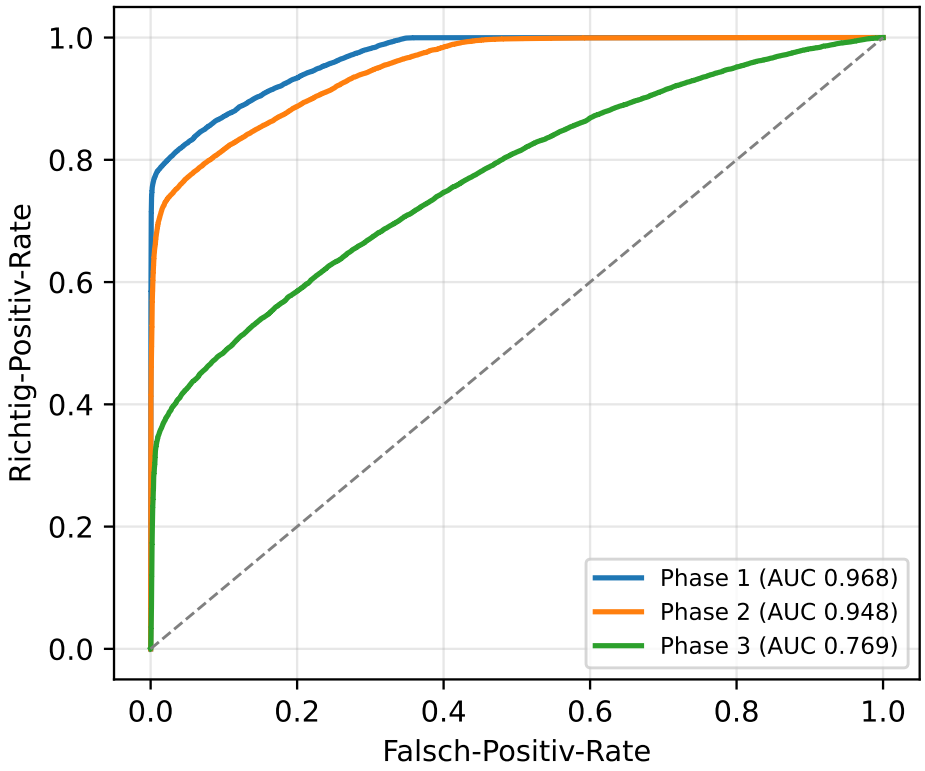


Figure 5.4.: ROC curves of the three phases for binary in-domain classification on AV-Deepfake1M++.

Table 5.3.: Contextualization of binary video-level detection on AV-Deepfake1M++ (AUC). Published method values are from their publications.

Method	Split	Training Data	AUC
HOLA [31]	TestA	∼1.1 M	0.9991
KLASSify [18]	TestA	∼1.1 M	0.9278
Phase 1 (this work)	Validation	∼50,000	0.968
Phase 2 (this work)	Validation	∼50,000	0.948
Phase 3 (this work)	Validation	∼50,000	0.769

between KLASSify and competition winner HOLA. Both procedures also employ extensive pre-training not used in this work.

5.3.2. FakeAVCeleb

On FakeAVCeleb, multimodal procedures such as MRDF [35] and AV-LGNN [30] report AUC values in range of approximately 0.97 to 0.99. In-domain values achieved in this work at AUC near 1.0 are in same order of magnitude. However, due to strong class imbalance and lack of unified official split, these values are only limitedly comparable and little informative, as discussed when presenting Table 5.2.

More informative is consideration of FakeAVCeleb as target dataset of cross-dataset evaluation. Named procedures trained on FakeAVCeleb and evaluated on it, so their values depict in-domain case. In the present work, Phase-2 model trained on AV-Deepfake1M++ achieves fusion AUC of 0.877 on FakeAVCeleb without seeing a single video of this dataset during training (see Table 5.4). The distance to in-domain trained procedures illustrates the cost of transfer to unknown manipulation techniques. Simultaneously, cross-dataset consideration answers a different and for practical deployment more relevant question than in-domain comparison since real manipulations rarely match training data distribution.

5.3.3. DeepSpeak v2.0

For DeepSpeak v2.0, only few comparable results exist since dataset was recently published and existing works sometimes use different evaluation protocols such as leave-one-out splitting by manipulation method [4]. Direct numerical comparison is therefore only partially possible. However, in-domain AUC achieved in this work of approximately 0.97 to 0.98 shows all three architectures solve the task in-domain.

Dataset authors motivate its creation by noting evaluation datasets must keep pace with current generation procedures [4]. Cross-dataset results of this work support this assessment: Models trained on AV-Deepfake1M++ fall to chance level on DeepSpeak across all modalities (see Table 5.4) despite the same audio branch remaining informative when transferring to FakeAVCeleb. DeepSpeak thus proves most demanding of the three datasets, its difficulty becoming apparent only upon transfer.

Overall essential difference to cited literature is that the present work systematically investigates cross-dataset generalization across all dataset combinations and separately by modality while published procedures predominantly report in-domain results.

5.4. Cross-Dataset Results

To assess generalization, models trained on AV-Deepfake1M++ are evaluated on the respective other datasets and vice versa. Table 5.4 shows results of models trained on AV-Deepfake1M++, separated by video branch, audio branch, and fusion.

Table 5.4.: Cross-dataset generalization (AUC) of models trained on AV-Deepfake1M++. Video, audio, and fusion AUC on respective target dataset are given.

Training → Test	**Phase**	**Video**	**Audio**	**Fusion**
AVD (4-class) → FakeAVCeleb	Phase 1	0.511	0.872	0.727
	Phase 2	0.833	0.836	0.699
	Phase 3	0.492	0.847	0.743
AVD (binary) → FakeAVCeleb	Phase 1	0.740	0.719	0.823
	Phase 2	0.829	0.697	0.877
	Phase 3	0.525	0.677	0.657
AVD (binary) → DeepSpeak	Phase 1	0.455	0.473	0.453
	Phase 2	0.466	0.504	0.460
	Phase 3	0.586	0.485	0.536

When generalizing from AV-Deepfake1M++ to FakeAVCeleb, a recurring pattern emerges. In four-class formulation, audio branch generalizes well and achieves AUC values between 0.836 and 0.872 across all phases. Video branch behaves inconsistently: while it falls to chance level AUC around 0.49-0.51 in Phases 1 and 3, X3D-M-based video branch in Phase 2 generalizes with AUC of 0.833. Fusion in four-class formulation lies with values around 0.70-0.74 below the respective better individual branch.

Generalization to DeepSpeak is poor for all phases and modalities. All AUC values lie between 0.45 and 0.59, near chance. Unlike FakeAVCeleb, audio branch does not generalize here. This is because audio manipulations in DeepSpeak differ from those in AV-Deepfake1M++, so learned acoustic features are not transferable.

Table 5.5 shows the reverse direction: models trained on FakeAVCeleb, evaluated on AV-Deepfake1M++.

Table 5.5.: Cross-dataset generalization (AUC) of models trained on FakeAVCeleb, evaluated on AV-Deepfake1M++.

Phase	**Video**	**Audio**	**Fusion**
Phase 1	0.536	0.512	0.513
Phase 2	0.510	0.487	0.518
Phase 3	0.506	0.531	0.501

Models trained on FakeAVCeleb do not generalize to AV-Deepfake1M++. All AUC values lie around 0.50, at chance level. This result contrasts with high in-domain values on FakeAVCeleb from Table 5.2 and illustrates that training on small and heavily imbalanced dataset does not provide transferable features. Models learn specific properties of FakeAVCeleb, not general manipulation features.

Comparable picture emerges for models trained on DeepSpeak. For Phase 1, the model trained on DeepSpeak was applied to the other two datasets. Table 5.6 shows results.

Table 5.6.: Cross-dataset generalization (AUC) of Phase-1 model trained on DeepSpeak.

Training → Test	**Video**	**Audio**	**Fusion**
DeepSpeak → AV-Deepfake1M++	0.507	0.526	0.529
DeepSpeak → FakeAVCeleb	0.476	0.630	0.715

Transfer of DeepSpeak to AV-Deepfake1M++ lies for all modalities near chance. When transferring to FakeAVCeleb, fusion achieves AUC of 0.715, but associated UAR is approximately 51%. This is attributable to FakeAVCeleb's strong class imbalance, where predicting majority class gives high accuracy but no true separation capability. Again, model trained on single dataset learns there-specific manipulations and is not transferable to different generation procedures.

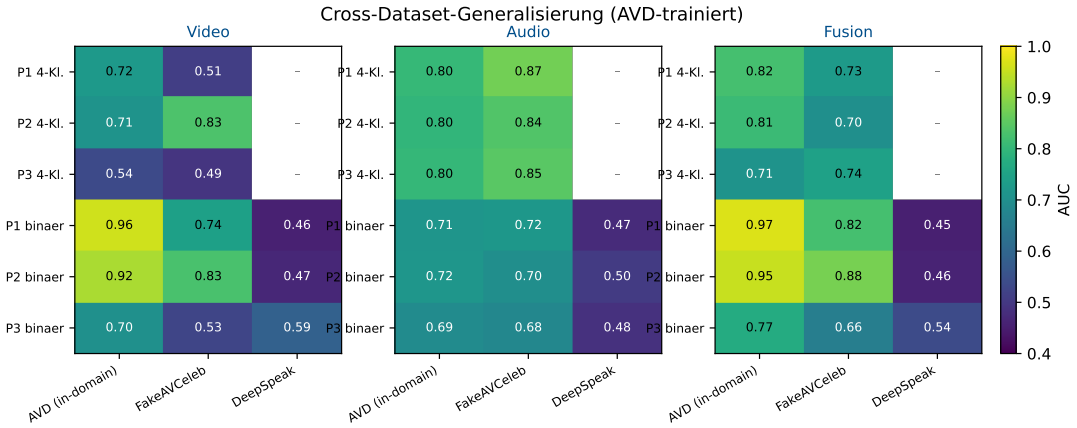


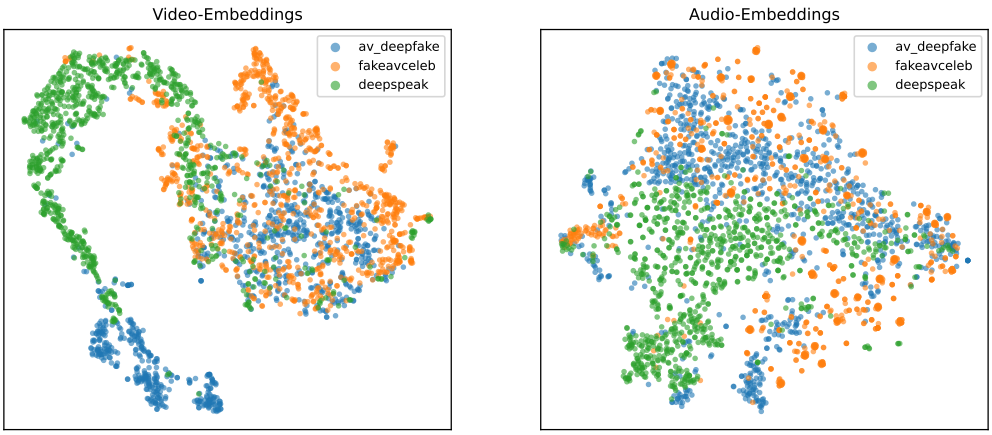
Figure 5.5.: Heatmap of fusion AUC across all training and test combinations. Diagonal shows in-domain performance; remaining cells show cross-dataset generalization.

5.5. Modality Analysis

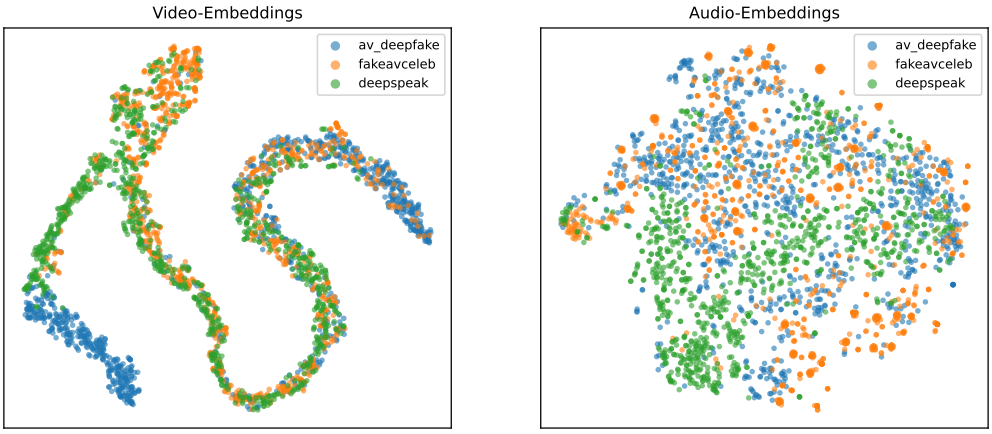
Separate evaluation by modality allows assessment of each branch's contribution individually. Table 5.7 shows in-domain results on AV-Deepfake1M++ separated by modality.

In-domain, video branch contributes largest portion to fusion in binary formulation. In Phase 1, video branch achieves AUC of 0.956 while audio branch reaches 0.712. Fusion improves result slightly to 0.968. In four-class formulation, the relationship partly reverses since audio branch is relevant for distinguishing audio-manipulated classes and achieves AUC values around 0.80.

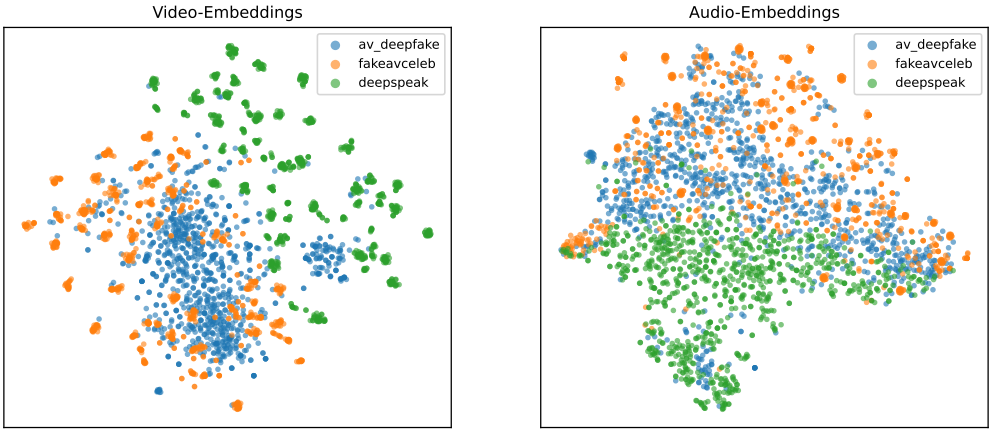
Comparison of in-domain and cross-dataset results reveals clear pattern. In-domain, video branch contributes strongly to performance. Cross-dataset, video branch collapses to chance level in most constellations while audio branch continues achieving AUC values around 0.85 when generalizing from AV-Deepfake1M++ to FakeAVCeleb. The video branch thus learns dataset-specific visual artifacts not transferable to other datasets, while HuBERT-based audio branch captures more general features.



(a) Phase 1 (ResNet-50 + BiLSTM)



(b) Phase 2 (X3D-M)



(c) Phase 3 (TimeSformer)

Figure 5.6.: t-SNE projection of video and audio embeddings of the three phases, colored by dataset. Audio embeddings of different datasets overlap more than video embeddings, illustrating better generalization of audio branch.

Table 5.7.: Modality-separated in-domain AUC on AV-Deepfake1M++.

Task	Phase	Video	Audio	Fusion
binary	Phase 1	0.956	0.712	0.968
	Phase 2	0.924	0.716	0.948
	Phase 3	0.702	0.690	0.769
4-class	Phase 1	0.725	0.802	0.824
	Phase 2	0.709	0.796	0.810
	Phase 3	0.536	0.799	0.709

Exception is X3D-M-based video branch in Phase 2, which generalizes cross-dataset on FakeAVCeleb with AUC of 0.833. This suggests spatiotemporal modeling of X3D-M captures more robust visual features than frame-based processing in Phase 1 or attention-based processing in Phase 3.

Another finding concerns fusion. In multiple cross-dataset constellations, fusion AUC lies below better individual branch AUC. When generalizing from AV-Deepfake1M++ to FakeAVCeleb in four-class formulation, audio branch in Phase 1 achieves AUC of 0.872 but fusion only 0.727. Concatenation cannot compensate poorly generalizing video branch but is pulled down by it. Cross-attention fusion in Phase 3 behaves similarly and does not solve this problem.

Multiple causes may account for this behavior. First, concatenation assigns fixed, learned-during-training weights to modalities. Classification head learns on training dataset how strongly to rely on individual dimensions of fused vector. Since video branch is informative in-domain (see Table 5.7), its features receive high weight. When video feature distribution shifts on foreign dataset, this weighting remains despite features having no information value there. Second, collapsed video branch provides not neutral noise but systematically shifted activations: Model assigns high confidence to patterns learned on training dataset that no longer correlate with classes on target dataset. Such activations act on classification head like reliable signal and thereby mask still-informative audio portion. t-SNE projection in Figure 5.6 supports this interpretation as video embeddings form clearly separated dataset clusters. Third, auxiliary losses apply exclusively to training dataset: They ensure both branches learn meaningful in-domain representations but lack mechanism estimating branch reliability at test time to incorporate into fusion.

That cross-attention fusion exhibits this behavior supports this explanation. While cross-attention computes modality interaction dynamically, attention weights are calculated from same shifted video features and cannot recognize their lacking information value. Both fusion strategies thus lack modality weighting dependent on reliability, as gated fusion would provide. This finding motivates the adaptive fusion approach sketched in Chapter 6.

5.6. Configuration Comparison

Beyond the three phases, individual configurations were systematically compared to determine their influence on performance. Two aspects were foremost: frame-sampling strategy in preprocessing and, for Phase 3, fusion strategy and input resolution.

5.6.1. Frame Sampling: Uniform versus Segment-Based

For visual modality, two sampling strategies were contrasted during training. First variant samples 16 frames evenly across entire video length. Second variant samples manipulated videos with 75% of frames deliberately from manipulated segments and distributes remaining frames evenly across rest of video. Temporal segment boundaries are available as annotation at training time and are used exclusively during training. At inference, both models use uniform sampling so both are evaluated under identical and realistic conditions. This comparison was conducted for Phase 1 in four-class formulation.

Table 5.8.: Influence of training sampling strategy in Phase 1 (4-class), evaluated with uniform sampling. Cross-dataset on FakeAVCeleb (test split).

	In-Domain (AVD)			**Cross (FakeAVCeleb)**		
Training	Video	Audio	Fusion	Video	Audio	Fusion
uniform	0.725	0.802	0.824	0.511	0.872	0.727
segment-based	0.815	0.779	0.764	0.869	0.979	0.774

The difference primarily affects video branch. With uniform training, video branch falls cross-dataset to AUC of 0.511, at chance level. If model trains segment-based, video branch achieves AUC of 0.869 cross-dataset. Even in-domain, segment-based trained video branch with 0.815 exceeds uniformly trained with 0.725.

This result indicates weak cross-dataset generalization of video branch is not inevitable but related to sampling strategy. With uniform sampling, video branch sees primarily authentic frames even for manipulated videos and learns dataset-specific global image features. When manipulated frames are deliberately included during training, video branch learns actual manipulation artifacts, which transfer better across datasets.

Uniform variant is nevertheless used for main experiments of all three phases since it requires no segment annotations and is uniformly applicable to all datasets. Segment-based variant shows however a concrete approach to deliberately improve video branch generalization and is taken up in Chapter 6 as starting point for future work.

5.6.2. Fusion Strategy and Resolution in Phase 3

For Phase 3, three configurations were compared on binary task of AV-Deepfake1M++. First uses cross-attention fusion at resolution 224 pixels, second same fusion at higher resolution 448 pixels with `timesformer-hr` model, and third concatenation fusion at 224 pixels.

Table 5.9.: Comparison of Phase-3 configurations on binary task of AV-Deepfake1M++ (Fusion).

Configuration	**ACC**	**UAR**	**AUC**
Cross-Attention, 224	66.7%	66.7%	0.739
Cross-Attention, 448 (HR)	68.5%	68.5%	0.752
Concatenation, 224	69.1%	69.1%	0.769

Results in Table 5.9 show concatenation with AUC of 0.769 performs best. Higher resolution improves cross-attention variant slightly from 0.739 to 0.752 but does not reach concatenation level. Additionally, 448-pixel

resolution increases computational cost: that run required approximately 55 hours compared to about 15 hours at 224 pixels, nearly four-fold, while achieving lower performance than concatenation. Since Phase 3's intended cross-attention fusion does not improve performance versus simple concatenation, concatenation variant is used as representative result for binary task and for cross-dataset evaluation. This substitution affects only binary task on AV-Deepfake1M++ and its cross-dataset transfers: All four-class results of Phase 3 plus DeepSpeak run use cross-attention fusion described in Chapter 4.

5.6.3. Weighting of Auxiliary Losses

Total loss comprises fusion, video, and audio loss with weights $\alpha = 0.6$, $\beta = 0.2$, and $\gamma = 0.2$ according to Equation 4.1. This division follows principle that fused representation is actual target and thus receives highest weight, while both modality-specific losses serve as regularization. Symmetric choice of β and γ treats both modalities equally during training so neither is preferred.

Systematic comparison of different weighting combinations was not conducted in this work. Chosen configuration proved stable in preliminary experiments, particularly inclusion of modality-specific losses prevented model relying exclusively on dominant modality. Systematic investigation of weighting constitutes possible starting point for future work.

5.7. Interpretation

Results allow several conclusions. First concerns asymmetry between modalities. Video branch achieves high in-domain values but generalizes poorly. This results because visual manipulation artifacts are dataset-specific and depend on particular generation method. Audio branch generalizes better since HuBERT as self-supervised pre-trained model provides general speech representations not tuned to specific dataset. This audio branch generalization ends where audio manipulation techniques fundamentally differ, as in DeepSpeak case. That video branch weakness is not fundamental failure of visual modality shows configuration comparison in Section 5.6: same video branch with segment-based training achieves cross-dataset AUC of 0.869.

Second conclusion concerns fusion. Concatenation does not weight modalities adaptively. Falls video branch cross-dataset to chance level, it worsens fused representation instead of remaining neutral. Cross-attention fusion employed in Phase 3 does not solve this since it incorporates both modalities into shared representation without downweighting malfunctioning branch. In Section 5.5 discussed causes, the in-domain learned weighting of classification head and systematically shifted video activations, explain why fusion in multiple constellations falls below better individual branch.

Third conclusion concerns architectural comparison. Increasing complexity from Phase 1 to Phase 3 does not lead to better performance. On the contrary, Phase 1 achieves highest in-domain AUC on AV-Deepfake1M++ while Phase 3 with TimeSformer and cross-attention declines both in-domain and cross-dataset. Possible explanation is TimeSformer as data-hungry transformer architecture could not be adequately adapted with approximately 50,000 training videos while convolutional encoders of Phases 1 and 2 manage better with this data amount. Content-driven manipulations of AV-Deepfake1M++ often affect only brief segments whose artifacts may be washed out by averaging over entire video in attention. Configuration comparison in Section 5.6 supports this: Neither higher input resolution nor cross-attention fusion improves Phase 3 performance, so weakness lies in video encoder itself and not in single configuration.

In summary, results show generalization of multimodal deepfake detectors is primarily carried by audio branch while video branch remains dataset-specific. Fusion strategy cannot balance this asymmetry, and more complex architectures under given conditions do not lead to better performance. These findings form basis for conclusions and future work approaches discussed in following chapter.

6. Conclusion and Outlook

6.1. Summary

This thesis has developed a multimodal system for video-level deepfake detection and systematically compared three architectural variants. Starting point was observation that existing detection procedures often operate unimodally or purely in binary, rarely investigate cross-dataset generalization, and lack controlled comparison of different architectures under identical conditions.

To address these gaps, three phases were developed differing exclusively in video encoder and fusion strategy while HuBERT-based audio branch and training procedure remained constant. Phase 1 combined ResNet-50 with BiLSTM, Phase 2 used X3D-M, and Phase 3 employed TimeSformer with cross-attention fusion. All models were trained in binary and, where datasets permitted, in four-class formulation on AV-Deepfake1M++, FakeAVCeleb, and DeepSpeak v2.0 and evaluated across all dataset combinations.

Results showed the pipeline functions reliably in-domain and achieves AUC values up to 0.968 on AV-Deepfake1M++ in binary formulation. Central finding concerns cross-dataset generalization. While audio branch remains informative on foreign datasets, video branch falls in most cases to chance level. This asymmetry was observed across all phases. Concatenation fusion could not compensate poorly generalizing video branch but was impaired by it. Cross-attention fusion in Phase 3 did not resolve this behavior. Higher complexity of Phase 3 did not pay off: both in-domain and under transfer it fell behind simpler approaches.

6.2. Limitations

This thesis is subject to several restrictions. Models were trained on single datasets each. Joint training across multiple datasets was not performed, so observed weak cross-dataset generalization may also result from this limitation.

Uniform frame sampling was used for main experiments. As shown in Section 5.6, sampling strategy influences video branch generalization, so reported video results are not to be understood as upper limit of achievable. Additionally, frame count was fixed to 16 and audio signal length to four seconds. Videos with different lengths or manipulations outside this window may be incompletely captured.

TimeSformer was trained with approximately 50,000 training videos. Since transformer architectures typically require large data amounts, it cannot be ruled out that Phase 3's subpar performance results from limited data amount. Furthermore, DeepSpeak was trained exclusively in binary, and FakeAVCeleb is heavily imbalanced, which limits result comparability between datasets. Finally, the thesis restricts itself to video-level classification so no statement about temporal position of manipulation is made.

6.3. Ethical Aspects

Deepfake detection is a dual-use tool. On one hand it serves to protect against misinformation and identity misuse; on the other, insights into detection procedures can be used to intentionally make manipulations harder to detect. A detection system should therefore be understood not as sole decision basis but as supporting tool within overall forensic assessment.

Used datasets contain biometric data in form of faces and voices of real people. When processing such data, applicable usage terms and data protection requirements must be observed. For instance, FakeAVCeleb includes video material of celebrities without their consent serving as basis for manipulations. Furthermore, misclassifications have consequences for affected people. False classification of authentic video as manipulation can be as harmful as undetected forgery. Since training data composition influences detection performance, there is risk that detector performs worse for certain person groups if underrepresented in training data.

6.4. Outlook

Results suggest several approaches for future work.

Most obvious would be improvement of fusion strategy. Since fusion is impaired by poorly generalizing video branch, adaptive fusion would be sensible that weights modalities depending on reliability. Such gated fusion could downweight malfunctioning branch instead of incorporating it equally into shared representation.

Furthermore results suggest training across multiple datasets. Present findings indicate training on single dataset leads to dataset-specific visual features. Joint training across multiple datasets could force video branch to learn more general features, thereby improving generalization.

Also remaining open is temporal localization. Present thesis deliberately restricted itself to video-level classification. Extension to segment-level localization would allow determining not only presence but also temporal position of manipulations. This is relevant for forensic applications requiring precise localization of manipulated sections.

Finally, investigation of TimeSformer with larger training amount could clarify whether its subpar performance results from limited data. Transformer architectures typically benefit from large data amounts so training on full AV-Deepfake1M++ dataset could change Phase 3 performance.

A. Supplementary Results and Configuration

A.1. Complete Cross-Dataset Results

Following tables compile all training-test dataset combinations each separated by video branch, audio branch, and fusion. AUC of best model is given. Diagonal, training and test on same dataset, corresponds to in-domain case. All values in this appendix derive from evaluation of saved best checkpoint on respective split. For FakeAVCeleb as target dataset, test split is used while in-domain values in Table 5.2 refer to validation split of best epoch. This results in minor decimal differences between both representations.

Table A.1.: Models trained on AV-Deepfake1M++, evaluated on all datasets (AUC).

Phase	Task	Test Dataset	Video	Audio	Fusion
Phase 1	binary	AV-Deepfake1M++	0.956	0.712	0.968
		FakeAVCeleb	0.740	0.719	0.823
		DeepSpeak	0.455	0.473	0.453
	4-class	AV-Deepfake1M++	0.725	0.802	0.824
		FakeAVCeleb	0.511	0.872	0.727
Phase 2	binary	AV-Deepfake1M++	0.924	0.716	0.948
		FakeAVCeleb	0.829	0.697	0.877
		DeepSpeak	0.466	0.504	0.460
	4-class	AV-Deepfake1M++	0.709	0.796	0.810
		FakeAVCeleb	0.833	0.836	0.699
Phase 3	binary	AV-Deepfake1M++	0.702	0.690	0.769
		FakeAVCeleb	0.525	0.677	0.657
		DeepSpeak	0.586	0.485	0.536
	4-class	AV-Deepfake1M++	0.536	0.799	0.709
		FakeAVCeleb	0.492	0.847	0.743

A.2. Hyperparameters

Table A.4 lists training parameters used uniformly for all three phases. Variations between phases concern exclusively video encoder and released layers.

Table A.2.: Models trained on FakeAVCeleb (4-class), evaluated on FakeAVCeleb (test split) and AV-Deepfake1M++ (AUC).

Phase	Test Dataset	Video	Audio	Fusion
Phase 1	FakeAVCeleb	0.994	1.000	0.9995
	AV-Deepfake1M++	0.536	0.512	0.513
Phase 2	FakeAVCeleb	0.992	1.000	0.9993
	AV-Deepfake1M++	0.510	0.487	0.518
Phase 3	FakeAVCeleb	0.988	1.000	0.989
	AV-Deepfake1M++	0.506	0.531	0.501

Table A.3.: Phase-1 model trained on DeepSpeak (binary), evaluated on all datasets (AUC).

Test Dataset	Video	Audio	Fusion
DeepSpeak	0.937	0.966	0.980
AV-Deepfake1M++	0.507	0.526	0.529
FakeAVCeleb	0.476	0.630	0.715

Table A.4.: Training configuration.

Parameter	Value
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning Rate Video/Audio	$3 \cdot 10^{-5}$
Learning Rate Fusion/Head	$1 \cdot 10^{-4}$
Weight Decay	0.01 (Video/Audio), 0.05 (Fusion)
Scheduler	Cosine with Warmup (800 steps)
Batch Size	32
Gradient Accumulation	2 (effectively 64)
Epochs	20
Gradient Clipping	0.5
Mixed Precision (AMP)	enabled
Label Smoothing	0.05
Auxiliary Weights	$\alpha = 0.6$, $\beta = 0.2$, $\gamma = 0.2$
Frames per Video	16
Image Resolution	224 (Phase 3 additionally 448)
Audio	16 kHz, 64,000 Samples (4 s)
Dropout	0.3
BatchNorm	frozen (eval mode)
Unfreezing ResNet-50	layer2, layer3, layer4
Unfreezing X3D-M	Blocks 3–4
Unfreezing TimeSformer	Layers 8–11
Unfreezing HuBERT	last 4 of 12 encoder layers

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